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A spatially distributed physical model for dynamic simulation of ventilated agro-material in bulk storage facilities

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ABSTRACT

To maintain a high quality and limit storage losses of agro-materials in bulk storage, the dynamic interaction between climate in the facility and product needs to be better understood. In this paper, we present a (2-D) spatially distributed, dynamic full-scale bulk storage model. Also, model calibration and validation results using data from a full-scale potato storage facility are presented. The model predicts convection and diffusion of heat, mass and carbon dioxide as well as the heat, mass and carbon dioxide transfer between bulk and air, respiration and evaporation of the food product and the natural and forced convection in the storage facility. In the CFD model two actuators, a moving hatch and a fan, were successfully implemented. The validated model was used as a tool to investigate the cooling process within the full-scale storage facility. When applying a combination of under- and overpressure instead of the conventionally applied overpressure, a reduction of the temperature gradient in vertical direction and an increased temperature gradient of the temperature gradient in horizontal direction was obtained. A reduction on this horizontal temperature gradient was accomplished by adding an air channel near the roof.

1. Introduction

Maintaining product quality of agro-materials is essential for the production and supply of excellent quality food products. Product quality is affected at each link in the food chain, hence also during storage (Jongen and Meulenberg, 2005). Throughout this paper the potato is used as an example of agro-material. During the storage process, the quality of the potato can change as a result of, for instance, internal temperature-dependent processes such as dormancy or cold-induced sweetening. In addition, processes in the product and at the product-air interface, such as respiration and evaporation, also affect the product quality. These processes determine, in particular, weight losses of the product and the level of carbon dioxide around the product. Hence, as the environment in and around the storage facility affects product quality, climate control is massively applied in bulk storage facilities for agro-materials.

Climate control in bulk storage facilities has improved in the past decades. Optimisation of ventilation strategies is only proceeding slowly, however, because of a lack of knowledge on the interaction between climate, a wide range of different and season-dependent agro-materials and the limited quality evaluation at the end of the season. To increase our understanding of the dynamics and interactions of a full-scale storage process and speed up the optimization of ventilation strategies, computational fluid dynamics (CFD) simulations could be of great help.

To perform CFD simulations, a spatially distributed mathematical model that captures all relevant physical, physiological and actuator processes should be build. In the case of frying potato bulk storage, temperature, moisture content and carbon dioxide are important processes that directly influence the product quality. The temperature, moisture content and carbon dioxide can be controlled by forced ventilation of (fresh) air through the storage facility. As generally known, the temperature and moisture content of a product has influence on all kinds of processes, like respiration, dormancy, rot, infection or sweetening. The moisture content has also a direct influence on the pressure bruising (Castleberry and Jayanty, 2012). Carbon dioxide levels affect the sugar content of the potato (see e.g. Mazza and Siemens, 1990; Daniels-Lake, 2013). Therefore, a CFD model that contains temperature, moisture content, carbon dioxide balances and actuators dynamics, of an appropriate air distribution is desirable.

In the literature both physically-based models (see e.g. Ofoli and Burgess, 1986; Verdijck et al., 1999; Jia et al., 2001; Nahor et al., 2004; Lukasse et al., 2007; Dehghannya et al., 2011) and empirical models based on physiological experiments (e.g. Tijskens and Polderdijk, 1996;

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Nomenclature

Subscript

a	air
p	produce

Parameters

α_c	heat transfer coefficient $[\frac{W}{m^2\,K}]$
κ	permeability $[m^2]$
λ	heat conduction coefficient $[\frac{W}{m\,K}]$
μ	dynamic viscosity [Pa s]
Ω	domain
ρ	density $[\frac{kg}{m^3}]$
τ	Cauchy stress
ε	porosity
θ	parameter vector
∇	vector differential operator nabla
CO_2	carbon dioxide concentration [ppm]
C	specific heat $[\frac{J}{kg\,K}]$
C_2	internal resistance $[\frac{1}{m}]$
$\mathbb{D}$	diffusion coefficient $[\frac{m^2}{s}]$
d_p	diameter of produce [m]
E_{CO_2}	respiratory CO_2 per unit of energy $[\frac{kg}{J}]$
E_{resp}	respiration mass per unit of energy $[\frac{kg}{J}]$
F_m	Potato surface $[\frac{m^2\,produce}{m^3\,bulk}]$
F_{eb}	external body force $[\frac{Pa}{m}]$
G_1	gain parameter
g	gravity $[\frac{m}{s^2}]$
h_{fg}	enthalpy of vaporisation $[\frac{J}{kg}]$
K_{evap}	product evaporation coefficient $[\frac{kg}{m^2\,s}]$
M	molar mass $[\frac{kg}{mol}]$
P	pressure [Pa]
R	gas constant $[\frac{J}{mol\,K}]$
$Resp$	product respiration rate $[\frac{W}{kg}]$
r_1	product respiration parameter $[\frac{W}{kg\,K}]$
r_2	product respiration parameter $[\frac{W}{kg\,K^2}]$
r_3	product respiration parameter $[\frac{W}{kg\,K^3}]$
t	time [s]
T	temperature [K]
$\boldsymbol{v}$	velocity vector $[\frac{m}{s}]$
X	moisture content $[\frac{kg}{kg}]$
X_s	saturated vapour concentration $[\frac{kg\,water}{kg\,air}]$

Hertog et al., 1997; Peppelenbos et al., 1996; Burg, 2004; Florkowski et al., 2014) are available for potato storage. Research related to CFD simulation of storage facilities can be found in the literature as well (see e.g. Kondrashov, 2000; Xu et al., 2002; Nahor et al., 2005; Chourasia and Goswami, 2007; Delele et al., 2009; Moureh et al., 2009; Delele et al., 2012; Ambaw et al., 2013; Hong et al., 2017). A calibrated and validated physically-based model that includes temperature, moisture content, carbon dioxide for both the air and product domain, and hatch and fan dynamics, as can be found in most storage facilities, is lacking.

The objective of the paper is to define, calibrate and validate a dynamic, spatially distributed model of the above mentioned processes in a full-scale potato storage facility for scenario studies. The proposed model in this paper describes the physical transport phenomena of both heat and mass. The flow of air is modelled by using the momentum balance and continuity equation. The complete dynamic model of the actual potato bulk storage facility with actuators (fan and hatches) with their corresponding air flows, was implemented in a CFD simulation tool. The CFD simulations were successfully used in the calibration and validation of the model. In a case study, the validated model was used to investigate a scenario with switching over- and underpressures, instead of the conventional applied overpressure.

In a conventional bulk storage facility overpressure is applied in the pressure chamber (left part in domain Ω_{a3} in Fig. 1), such that by means of forced convection the air streams from the bottom to the top of the bulk (blowing). From a construction point of view, this is an efficient way to apply a homogeneous distribution of air in the horizontal direction. However, from practise and the results presented in Grubben and Keesman (2017), controlling the temperature at the top of the bulk cannot be reached in short time periods, and temperature and moisture gradients in vertical direction are seen. Hence, to control the top of the bulk more efficiently, the air should also stream from the top to bottom of the bulk. This can be realised by applying underpressure in the pressure chamber (suction). As conventional storage facilities are not constructed to supply a homogeneous distribution in suction mode, a CFD model will provide insight into the development of the climate and product quality in the storage facility using a combination of blowing and suction ventilation.

This paper is structured as follows. First, we set out the modelling procedure. Second, we show how we calibrated and validated the resulting CFD model, with experimental data from a full-scale potato

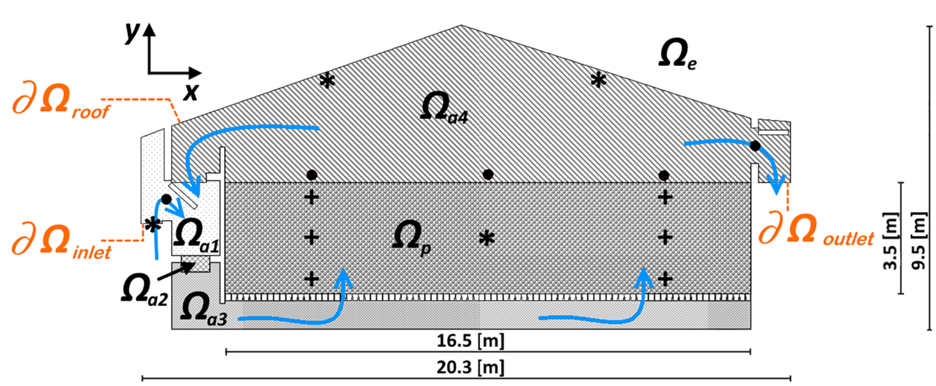

Fig. 1. 2-D configuration of the potato storage facility, with Ω_a the air domain, Ω_p the porous medium domain and Ω_e the environment domain. The climate is controlled by mixing and ventilating air from the outside Ω_{a1} and the inside Ω_{a4}, via fan Ω_{a2} and air channel Ω_{a3} through the porous medium Ω_p. The air can leave the facility or can be recirculated. The symbols •, $*$ and + represent measuring points for air velocity, and for temperature, moisture content and carbon dioxide. The inlet, outlet and roof boundaries are denoted by $\partial\Omega_{inlet}$, $\partial\Omega_{outlet}$ and $\partial\Omega_{roof}$.

storage facility under active control of temperature, moisture content and carbon dioxide. The results of the calibration, validation and case study are given in Section 3. The results are discussed in Section 4.

2. Material and methods

This section presents a description of the full-scale potato storage facility including the actuators and sensors locations. Subsequently, the model assumptions are presented, followed by a mathematical description of the relevant processes. Next, the simulation and calibration procedure is described. Finally the case study is introduced.

2.1. Large scale storage facility

To obtain data for calibration and validation of the model, we monitored a full-scale potato storage facility in the Netherlands, (Flevoland) during the storage season of 2015–2016. The used variety was Ramos, that was grown for 173 days and the haulm was killed 14 days before harvesting. The potatoes were harvested on 6th October, and treated with 150 ml Grow-Stop ready (CIPC) per 1000 kg during storage filling, for sprout inhibition. The potatoes were stored in a bulk storage facility in a pile of 3.5 m high for 259 days. For the calibration and validation data from 13 to 14 March and from 6 to 7 April were collected, respectively. Fig. 1 depicts the facility with width of $x = 20.3$, height of $y = 9.5$, and a length of $z = 12.5$ m, which was storing around 500 tonnes of potatoes. For modelling purposes, we defined four different air domains Ω_{ai}, with $i = 1, \ldots, 4$, one porous domain Ω_p and an environment domain Ω_e. We placed sensors at several positions in the facility; see Fig. 1. Table 1 lists the specifications of the sensors and its positions. The considered 2-D plane in the middle of the facility, hence at $z = 6.25$ [m].

In the storage facility the temperature, air humidity and carbon dioxide level were controlled. The climate was controlled with only 2 actuators, namely the fan and the inlet and outlet hatches. When the temperature or carbon dioxide level exceeded a certain level, the hatches opened and through a forced convection, colder and fresh air entered the store. This was only done when the air humidity of the incoming air was in an acceptable range. When the temperature distribution exceeded a pre-specified level of heterogeneity, the fan homogenized the bulk with the hatches closed. From Fig. 1 the corresponding flows can be described in more detail. In domain Ω_{a1} air from the environment (Ω_e) can enter the store of can be mixed with air from within the facility using the hatch. This air is ventilated into the air channel (Ω_{a3}) by making use of a fan (Ω_{a2}), that creates an overpressure. From the air channel the air is ventilated through the perforated floor into the product domain (Ω_p). Once the air is ventilated through the product domain, the air can leave the facility or can be recirculated (Ω_{a4}). In general, the climate conditions are set by rules of thumb. For the Ramos cultivar the temperature was set at 8 degrees, the humidity around the 95% and the carbon dioxide level should never exceed 4000 ppm. The facility is operated with an overpressure over the bulk.

2.2. Model assumptions

In a dynamic simulation of a complete bulk storage facility, the model boundaries are determined by the physical borders of the storage facility. Disturbances in the ambient environment can influence the internal domains of the facility, through the borders and actuators, such as fan and hatches. For a full-scale storage facility, air domains and a porous media domain are distinguished (see Fig. 1).

As the model should be used to investigate ventilation strategies in full-scale potato facility, the relevant dynamics should be described by a spatially distributed model. Therefore, only the processes that contribute most to short-term changes in product quality are modelled. The model describes the dynamics of temperature, moisture content, carbon dioxide in air and product domain, and includes a flow model. In the CFD implementation also two types of actuators, hatches and fan, are included.

The following assumptions with respect to bulk potato storage are part of our model:

- The interior of the potato and the potato skin are modelled by a combined diffusion coefficient.
- The distribution of the potatoes in the bulk is uniform and is modelled as a large spherical packed beds approximation (Sman, 2002). This simplification also allows a two-phase modelling approach, which implies that it is not necessary to draw the actual physical potatoes and knowing the ratio between air ($\varepsilon_a = \varepsilon$) and product ($\varepsilon_p = 1 - \varepsilon$) suffices (see Fig. 2).
- The quality of the potato is defined in terms of temperature-dependent evaporation and respiration that both drive product weight losses. Furthermore it is assumed that the product weight loss is equal to the product moisture loss, thus neglecting weight loss due to other mass losses.
- Condensation is negligible.
- No internal rotting is present. (If a percentage of rot is observed during harvesting or during the intake of the product, the average rates of rotting-related processes, such as respiration of the potatoes, should be adjusted in the model.)

2.3. Spatially distributed model

In the following, we propose a dynamic and spatially distributed model, based on the work of Sman (2002), Lukasse et al. (2007), Chourasia and Goswami (2007) and Kondrashov (2007), to describe the spatially distributed physically and physiologically related processes.

In the model, each individual process variable is described by a partial differential equations (PDE) and called a state equation. The processes in the facility take place in two different domains (Ω_{ai} and Ω_p), which are linked to each other. The air domains (Ω_{ai}) contain both air and the control variables, defined as the ventilator pressure and angular displacement of the moving hatches. Hence, the air domains contain only state equations related to physical phenomena. The porous domain (Ω_p) is described by a two-phase model, including the product and air domain. The state equations describe physical phenomena and physiologically related processes in the porous domain. These state equations follow from the laws of conservation of energy and mass. The air flow is described by the law of conservation of momentum and the continuity equation.

The law of conservation of energy describes the temperature (T [K]) change in the porous medium (Ω_p) and in the air domains (Ω_{ai}, $i = 1, \ldots, 4$). For the porous medium and air domain, the following three state equations describe the temperature profiles:

Table 1
Overview of measuring points.

Point	Domain	Sensor	Specifications	Accuracy
•	Ω_{a1}	v	Anemometer[a]	±0.1 [m/s]
•	Ω_{a4}	v	Anemometer[a]	±0.1 [m/s]
*	Ω_{a1}	T_a	Pt100[a]	±0.1 [°C]
		X_a	HC101[a]	±3 [%]
		CO_2	Transmitter[a]	±50 [ppm]
*	Ω_{a4}	T_a	Pt100[a]	±0.1 [°C]
*	Ω_p	T_p	Pt100[a]	±0.1 [°C]
		X_p	Load cell[b]	±5 [g]
		CO_2	Transmitter[a]	±50 [ppm]
+	Ω_p	T_p	Pt100[a]	±0.1 [°C]

[a] Manufacturer: E + E Elektronink, Austria.
[b] Manufacturer: AllScales Europe, Netherlands

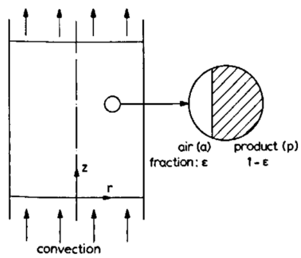

Fig. 2. Two-phase modelling approach for porous medium (Beukema et al., 1982).

$$\frac{\partial T_a}{\partial t} + \frac{1}{\varepsilon_a}\boldsymbol{v}\cdot\nabla T_a - \nabla\left(\frac{\lambda_a}{\rho_a C_{pa}}\nabla T_a\right) = \frac{\alpha_c F_m}{\rho_a C_{pa}\varepsilon_a}(T_p - T_a) \text{ in } \left(0,\ T\right] \times \Omega_p \tag{1}$$

$$\frac{\partial T_p}{\partial t} - \nabla\left(\frac{\lambda_p}{\rho_p C_p}\nabla T_p\right) = -\frac{\alpha_c F_m}{\rho_p C_p \varepsilon_p}(T_p - T_a) + \frac{Resp}{C_p} - h_{fg} K_{evap}\frac{F_m}{\rho_p C_p \varepsilon_p}\left(X_s - X_a\right) \text{ in } \left(0,\ T\right] \times \Omega_p \tag{2}$$

$$\frac{\partial T_a}{\partial t} + \boldsymbol{v}\cdot\nabla T_a - \nabla\left(\frac{\lambda_a}{\rho_a C_{pa}}\nabla T_a\right) = 0 \text{ in } \left(0,\ T\right] \times \Omega_{ai} \tag{3}$$

Here, $T_a \coloneqq T_a(x, y)$ is the air temperature as a function of the spatial coordinates, $T_p \coloneqq T_p(x, y)$ is the potato temperature, $\nabla \coloneqq \left[\frac{\partial}{\partial x}\ \frac{\partial}{\partial y}\ \frac{\partial}{\partial z}\right]^T$ the vector differential operator nabla, with $z = 0$, $\frac{\partial}{\partial z} = 0$, $\frac{\partial^2}{\partial z^2} = 0$, $\boldsymbol{v} = [v_x\ v_y\ 0]^T$ the velocity vector [m/s], X_a the moisture content of the air [kg/kg] and ε the porosity ($\varepsilon_a = 0.36$), h_{fg} the enthalpy of vaporisation ($2490\cdot10^3$ [J/kg]), K_{evap} the product evaporation coefficient [kg/(m^2 s)], and estimated during a calibration step. Furthermore, λ is the heat conduction coefficient [W/(m K)], ρ the density [kg/m^3] and C the specific heat (C_{pa} and C_p for air and product, respectively) [J/(kg K)]. For air these coefficients are calculated dynamically by COMSOL, for the potato these are taken as 0.4, 1014 and 3516 respectively. At last, F_m represents the specific surface of the mount [m^2/m^3], $Resp$ the respiration [W/kg], X_s the average vapour concentration around the potato [kg/kg] and α_c the heat transfer coefficient [W/(m^2 K)] and these are explained below.

The first term on the right-hand side of the PDEs for T_a and T_p in the porous domain in Eqs. (1) and (2) characterizes the heat transfer between air and product. In this term, the specific surface of the bulk F_m stands for potato surface per cubic meter in the bulk (Lukasse et al., 2007), which is given by:

$$F_m = \frac{\pi d_p^2}{(1/6)\pi d_p^3}\varepsilon_p \tag{4}$$

Here, d_p the potato diameter (0.06 [m]). The second and third reaction term on the right-hand side of the PDE for T_p in Eq. (2) define the respiration and evaporation, respectively. The product respiration rate $Resp$ in Eq. (2) is given by the following equation (Lukasse et al., 2007):

$$Resp = r_1 + r_2(T_p - 273.15) + r_3(T_p - 273.15)^2 \tag{5}$$

Here, r_1, r_2 and r_3 are the product respiration parameters ([W/(kg K)], [W/(kg K^2)] and [W/(kg K^3)] respectively), and are estimated during the model calibration step. For the evaporation, the latent heat of evaporation h_{fg} and the product evaporation coefficient K_{evap} are taken as constants (Lukasse et al., 2007). The evaporation is influenced by the equilibrium with the moisture content in air, which is temperature-dependent. The average vapour concentration at the potato surface is approximated by the Antoine equation for water vapour (Lukasse et al., 2007):

$$X_s = \frac{e^{\left(23.5 - \frac{3991}{T_{skin} - 39.15}\right)}}{\frac{1}{M_w}\rho_a R T_a} \tag{6}$$

Here, M_w is the molar mass of water (0.018 [kg/mol]), R the gas constant (8.31446 [J/(mol K)]) and T_{skin} the temperature of the skin implemented likewise as in Lukasse et al. (2007). The heat transfer coefficient α_c represents the transfer of heat between the potatoes and the air (Kondrashov, 2007), and is given by:

$$\alpha_c = \alpha_{c1} d_p + \frac{\alpha_{c2} v^{0.67}}{d_p^{0.33}} \tag{7}$$

Here, $\alpha_{c1} = 0.05$ [W/(m^3 K)] and α_{c2} [J/(m^2 K)] is estimated in a calibration step.

The law of conservation of mass describes the moisture profiles for the porous domain (X_p [kg/kg product], X_a [kg/kg air]) and air domain (X_a [kg/kg air]). Similarly as for the temperature-related states, the moisture profile in the porous medium and air domains is modelled with three different state equations, as follows:

$$\frac{\partial X_a}{\partial t} + \frac{1}{\varepsilon_a}\boldsymbol{v}\cdot\nabla X_a - \nabla(\mathbb{D}_a \nabla X_a) = K_{evap}\frac{F_m}{\rho_a \varepsilon_a}\left(X_s - X_a\right) \text{ in } \left(0,\ T\right] \times \Omega_p \tag{8}$$

$$\frac{\partial X_p}{\partial t} - \nabla(\mathbb{D}_p \nabla X_p) = -K_{evap}\frac{F_m}{\rho_p \varepsilon_p}\left(X_s - X_a\right) - E_{resp} Resp \text{ in } \left(0,\ T\right] \times \Omega_p \tag{9}$$

$$\frac{\partial X_a}{\partial t} + \boldsymbol{v}\cdot\nabla X_a - \nabla(\mathbb{D}_a \nabla X_a) = 0 \text{ in } \left(0,\ T\right] \times \Omega_{ai} \tag{10}$$

Here, $X_a \coloneqq X_a(x, y)$ is the moisture content in the air, and $X_p \coloneqq X_p(x, y)$ the moisture content of the potato. The first reaction term on the right-hand side of the PDEs for X_a and X_p in the porous domain in Eqs. (8) and (9) captures the moisture exchange between air and product. The second term on the right-hand side of the PDE for X_p describes the moisture production due to respiration. This respiration is temperature-dependent (Eq. (5)), with the respiration mass per energy ($E_{resp} = 2.5\cdot10^{-8}$ [kg/J]) taken as a constant. Here, $\mathbb{D}_a$ the diffusion coefficient in air ($2.2\cdot10^{-5}$ [m^2/s]) and $\mathbb{D}_p$ the diffusion coefficient in the potato ([m^2/s]) and is taken as (Xu and Burfoot, 1999):

$$\mathbb{D}_p = 5.1\cdot10^{-3}\cdot e^{\left(-13.5X_p\right)\left[-3151.5\left(\frac{1}{T_p} - \frac{1}{313.15}\right)\right]} \tag{11}$$

The CO_2 concentration is modelled with the aid of two state equations for $CO_{2,a}$ [%], in the porous medium and air domain, using the law of conservation of mass:

$$\frac{\partial CO_{2,a}}{\partial t} + \frac{1}{\varepsilon_a}\boldsymbol{v}\cdot\nabla CO_{2,a} - \nabla(\mathbb{D}_{CO2,a}\nabla CO_{2,a}) = E_{CO2} Resp \text{ in } \left(0,\ T\right] \times \Omega_p \tag{12}$$

$$\frac{\partial CO_{2,a}}{\partial t} + \boldsymbol{v}\cdot\nabla CO_{2,a} - \nabla(\mathbb{D}_{CO2,a}\nabla CO_{2,a}) = 0 \quad \text{in } \left(0,\ T\right] \times \Omega_{ai} \tag{13}$$

Here, $CO_{2,a}$;=$CO_{2,a}(x, y)$ is the CO_2 content in the air, $\mathbb{D}_{CO2,a}$ the diffusion coefficient ($1.6\cdot10^{-5}$ [m^2/s]). The production term of the CO_2 in the porous domain is driven by the respiration of the potato (5). The respiratory CO_2 concentration per energy E_{CO2} ([kg/J]) is taken as a constant (Lukasse et al., 2007), and estimated in a calibration step.

Air flow is described by an incompressible, Reynolds Averaged Navier Stokes (RANS) with Algebraic yPlus turbulence model. Additional, a volume force in domain (Ω_{a2}) simulates the fan and an aerodynamic resistance is taken into account in the porous domain (Ω_p) (Chourasia and Goswami, 2007). This porous domain experiences small (natural) as well as large (forced) convective flows. In the natural convection case, the airflow is proportional to the pressure drop and

Darcy's law is valid for small air flows. In the case of forced convection, the air in the porous medium has a relative low viscosity. Owing to this low viscosity, the inertial forces in the Reynolds number have a dominant effect, which is described by the Forchheimer resistance. Hence, the total external body force (F_{eb} [Pa/m]) is a combination of the Darcy coefficient and a term that covers the inertial force caused by the kinetic energy of the air.

$$F_{eb} = \underbrace{-\frac{\mu}{\kappa}v_i}_{\text{Darcycoefficient}} \underbrace{-C_2\frac{1}{2}\rho_a\bar{v}v_i}_{\text{inertialterm}} \tag{14}$$

Here, μ is the dynamic viscosity [Pa/s], and calculated dynamically by COMSOL, κ is the permeability [m^2] and C_2 an internal resistance factor [1/m]. The permeability is a property that indicates the ability of the air to flow through the porous domain. The resistance term C_2 is an internal resistance factor. Ergun (1952) has given general approximations for spherical packed beds. Chourasia and Goswami (2007) have expressed the permeability and internal resistance on the basis of Ergun's equation for the potato bulk as follows:

$$\begin{aligned} \kappa &= \frac{d_p^2\varepsilon_a^3}{c_1(\varepsilon_p)^2} \\ C_2 &= \frac{c_2\varepsilon_p}{d_p\varepsilon_a^3} \end{aligned} \tag{15}$$

Here, the parameters c_1 and c_2 were estimated in a calibration step.

2.4. Numerical simulation

CFD simulation software (COMSOL version 5.2a) was used for simulate and analyse the model, as defined in Eqs. (1)–(15). To simulate the movement of the hatches, we implemented a moving mesh. A moving mesh allows to describe a displacement of a spatial frame by automatic generating a new mesh when needed. A new mesh is needed as the spatial frame (hatch) disturbs the mesh quality during dynamic simulations. If the mesh quality reaches a certain level of inaccuracy, automatically a new mesh grid is created.

The flow model was decoupled from the product (quality-related) model in the calibration step in view of the temporal and spatial complexity of the facility. In the validation step, the complete model was simulated, however, including the dynamic behaviour of the fan and moving hatches.

2.4.1. Boundary conditions

For the storage facility simulation, with configuration as in Fig. 1, different boundary conditions related to the proposed states in Section 2.3 were used. All boundary conditions for the boundary of the facility, as shown in Fig. 1, are taken as no flux (Neumann) through the boundary for temperature, moisture content and carbon dioxide, with some exceptions for the three boundary conditions $\partial\Omega_{roof}$, $\partial\Omega_{inlet}$ and $\partial\Omega_{outlet}$. For the roof ($\partial\Omega_{roof}$) a Dirichlet condition is set for the temperature ($T_a = T_{roof}$). For the inlet ($\partial\Omega_{inlet}$) Dirichlet conditions were set for temperature ($T_a = T_{a,inlet}$), moister concentration ($X_a = X_{a,inlet}$) and carbon dioxide concentration ($CO_2 = CO_{2,inlet}$). For the outlet ($\partial\Omega_{outlet}$) Nuemann conditions, in the form of an outflow condition were used for temperature ($\frac{\lambda_a}{\rho_a C_{pa}}\nabla T_a = 0$), moister concentration ($\mathbb{D}_a\nabla X_a = 0$) and carbon dioxide concentration ($\mathbb{D}_{CO2,a}\nabla CO_{2,a} = 0$). For the flow model no slipping boundary conditions were used except for the inlet and outlet. For these boundaries ($\partial\Omega_{inlet}$ and $\partial\Omega_{outlet}$) both zero static pressure inlet- and outlet boundary condition were used.

2.5. Calibration

A calibration of the bulk storage model needs to be performed before the model can realistically simulate the behaviour of air and a specific agro-material. All unknown or uncertain model parameters that are product-specific or related to the climate should be estimated. We carried out the parameter estimation with data from measurements in a full-scale bulk storage facility for potatoes. To limit the number of parameters and thus the computation time, the system was decoupled into a fast (air) model and a slow (product) model.

In the first calibration step, we only took the flow properties of the complete bulk storage facility into account. The flow is governed by convection and diffusion, and we therefore estimated the parameters c_1, c_2 and the fan pressure P_{fan} first. The parameter estimates from the flow model simulations were then compared with data obtained at four locations in the facility over a period of 48 h. The data pertain to the air velocity in the inlet, outlet, and three points right above the bulk (see Fig. 1).

Next, the parameters of the product model, K_{evap}, r_1, r_2, r_3, α_{c2} and E_{CO_2} were estimated. In this second calibration step, the geometry was decoupled to limit computation time and only the porous domain (Ω_p) was taken into account. The product model simulation results were compared with data obtained in the facility (see Fig. 1) over a period of 48 h. Only the potato temperature T_p, moisture content of the potato X_p, and the carbon dioxide concentration CO_2 at the centre of the bulk were measured and used for the calibration. Once all parameters were estimated, we coupled the system again to analyse the validity of the complete dynamic CFD simulation model. In addition to the midpoint validation, also six other points in the bulk (as shown in Fig. 1 and Table 1) were used to validate the spatial distribution of the temperature.

For the model calibration, a range of parameter combinations was evaluated, using a parameter sweep. We defined the range of parameters on the basis of physical knowledge and after running some model simulations. This parameter sweep approach works more efficiently than a (non)-linear least squares optimisation method (LSQ) for this full-scale system. Unlike a LSQ method with potential local minima, a parameter sweep approach yields, for a predefined number of grid points, quantitative knowledge about the feasible parameter domain.

2.5.1. Flow model calibration

The exchange between climate and product in the bulk storage facility takes place by diffusion and natural convection, and is amplified by forced convection. Most of the diffusion rates are known from the literature, and we assumed that these rates hold for all bulk storage facilities. The air flow is defined by the RANS model, including a Darcy-Forchheimer resistance that describes the resistance of the porous medium; see Eq. (15). For the parameter sweep, we first calculated the physically possible range for parameters c_1, c_2 and P_{fan} on the basis of the experiments of Irvine et al. (1993).

Irvine et al. (1993) defined resistance factors for different clean and soiled potato species, and used a 0.95 m^3 measuring set-up. Sman (2002) used the clean potato data of Irvine et al. (1993) to prove, in particular that the Darcy-Forchheimer resistance is able to describe the bulk resistance for clean potatoes. However, in practice, soil contents between 2 and 5% are present. Hence, we first used the data of Irvine et al. (1993) to provide a first guess of the range for the parameters c_1 and c_2. Second, we carried out a parameter sweep to estimate the parameters c_1, c_2 and P_{fan} for the monitored full-scale bulk storage facility.

The available data from Irvine et al. (1993) comprise two different sets related to two bulk compositions. The first set is related to a bulk with a potato diameter of $dp = 65$ [mm] and a porosity of $\varepsilon_a = 0.36$, whereas the second set pertains to a potato diameter of $dp = 61$ [mm] and a porosity of $\varepsilon_a = 0.38$. Using CFD simulations of the complete flow field in the facility, we performed the parameter sweep for soil percentages of 0, 2.5, 4 and 6% in the bulk. The parameter sweep used the following intervals: $c_1 \in [100, 260]$ with $\Delta c_1 = 5$ and $c_2 \in [0, 20]$ with $\Delta c_2 = 0.1$, and $P_{fan} \in [80, 150]$ with $\Delta P_{fan} = 5$.

2.5.2. Product model calibration

Subsequently, we calibrated the product model on Ω_p only. The product model states are the temperature (T_p), moisture content (X_p) and carbon dioxide (CO_2) concentration. These states show relatively slow dynamics in comparison with air flow. For the porous domain, we calculated three respiration coefficients r_1, r_2, r_3 of the respiration term R, the coefficient of the heat transfer coefficient α_{c2}, the evaporation coefficient K_{evap}, the respiratory CO_2 concentration per energy E_{CO_2} and a gain parameter G_1 on the velocity ($\boldsymbol{v}_{eff} = G_1 \cdot \boldsymbol{v}$).

As described before, we used a parameter sweep instead of a formal LSQ optimisation method to estimate the parameters. For the product model literature values can be found (see Table 3). On base of the literature values three times a hypercube on three levels is performed. To determine the influence of changes in a parameter estimate on the output-weighted mean squared error (WMSE^x), with weights $w(x) = 1/var(x)$, we also calculated the sensitivity of WMSE^x with respect to a specific parameter θ_i. The normalised sensitivities, around the specific parameter estimate θ_i^*, were obtained with the following equation:

$$S_{\vartheta_i}^x = \frac{WMSE^x(1.1 * \theta_i^*) - WMSE^x(0.9 * \theta_i^*)}{0.2 * \theta_i^*} \times \frac{\theta_i^*}{WMSE^x(\theta^*)} \tag{16}$$

Here, x is T_p, X_p or CO_2, and a large sensitivity ($S_{\vartheta_i}^x$) implies a large influence of a specific parameter estimate (θ_i) on the weighted mean squared error related to state x, WMSE^x. This analysis makes it possible to determine the dominant parameters from the seven parameters we initially chose.

2.6. Case study

For (frying) potatoes, in general, a constant temperature and minimal weight loss is desirable. The temperature and temperature fluctuations have influence, on for instance, the respiration, sugar development (Hertog et al., 1997), ageing (Coleman, 2000) and dormancy (Struik and Wiersema, 2012; Muthoni et al., 2014). Weight loss has influence on the net yield at the end of the storage period, process ability of the potato and on pressure bruising defects. Since the potato is a living organism and produces heat, in practice, at least once a day the facility is ventilated with cool and fresh air. As shown in Fig. 1, the cool air is only ventilated into the storage by making use of an overpressure. Therefore, relative large temperature and weight loss gradients in vertical direction are obtained. As a result, the lower part of the bulk storage are exposed to colder and dryer air. This heterogeneity can cause condensation (Pringle et al., 2009), and a higher weight loss in the bottom part of the facility. Especially, the weight loss, in combination of the pressure in the bottom layer from the potatoes on top of it, are a cause of pressure flattening defects in the bottom layers.

In this case study (case study 1) the affect of a combination of over and underpressure is investigated for existing facilities. We hypothesize that a smaller gradient in vertical direction is obtained if a combination of over- and underpressure is applied. In the case study the simulation of the first cooling period, 8.6 to 10.0 [h] of the validation data set will be used. In the original case from 8.6 to 10.0 [h] an overpressure was applied. For the case study from 8.6 to 9.3 [h] an overpressure and from 9.3 to 10.0 [h] an underpressure is applied at the fan.

3. Results

This section first presents the results of the calibration of the flow and product model, followed by the validation of the complete CFD model. The simulations for validation with a data set of 36 h was performed on an Intel(R) Xeon(T) SPU Ed5-1650 0, 6 cores, 3.2 GHz, 24 GB RAM and took eventuality 36 h 06 m 40s. Finally, in Section 3.4 the results of the case study will be presented.

3.1. Flow model calibration

For unsoiled potatoes, a potato diameter of $d_p = 65$ [mm] and a porosity of $\varepsilon_a = 0.36$, Van der Sman (2002) derived values of $c_1 = 180$ and $c_2 = 3.6$, which are typical values for spherical products (Macdonald et al., 1979). This result corresponds to our findings, as shown in Table 2. The performed parameter estimation procedure resulted in a constant value for $c_1 = 180$ and a varying value for c_2 with $c_2 \in [3.6, 14.5]$.

Table 2
Forchheimer resistance.

Soil percentage	0	2.5	4	6
dp = 65 mm, $\varepsilon_a = 0.36$	$c_2 = 3.6$	$c_2 = 5.8$	$c_2 = 10.4$	$c_2 = 13.6$
dp = 61 mm, $\varepsilon_a = 0.38$	$c_2 = 4.2$	$c_2 = 6.9$	$c_2 = 11.3$	$c_2 = 14.5$

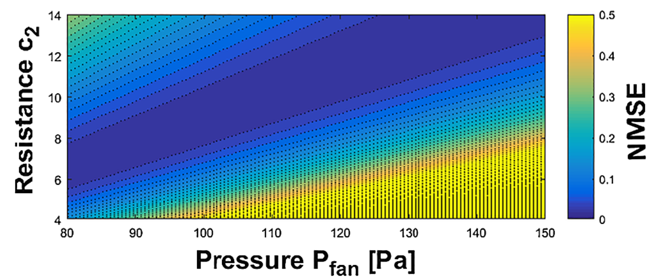

Fig. 3. Normalised mean squared error of the Darcy-Forchheimer resistance calibrated with data of a full-scale facility, with dp = 0.060 [m] and $\varepsilon_a = 0.360$.

The potato diameter d_p that we used was estimated as 60 mm with a porosity of $\varepsilon_a = 0.36$. The tare average of soil is 4.5%. Since these values are different from the data of Irvine et al. (1993), we needed to conduct a new parameter estimation step. From the first parameter estimation step (Table 2), we found that $c_1 = 180$. For the estimation of c_2 and P_{fan}, again a parameter sweep was performed. From the literature, we knew that the pressure P is between 80 and 150 [Pa] (Rastovski, 1987), and c_2 is between 4 and 14. Fig. 3 shows the result of the second parameter sweep.

In Fig. 3, a valley of parameter estimates with a small normalised mean squared error (NMSE) can be observed. This valley indicates a strong positive correlation between the estimates. The smallest NMSE ($1.7\ 10^{-6}$) occurs for the parameter estimates $\hat{c}_2 = 12.5$ and $\hat{P}_{fan} = 130$ [Pa].

3.2. Product model calibration

Table 3 lists the parameter estimates ($\hat{\theta}_i$) using the measured states and the original parameters as found in the literature (θ_{Lit}) (columns 3 and 2 respectively) and Fig. 4 depicts the corresponding calibration results.

Table 3
Product model parameter estimation. For the parameters that show a consirable influence on the corresponding states, the parameter sensitivities are presented in bold.

ϑ_i	$\vartheta_{i,Lit}$	$\hat{\vartheta}_i$	Sensitivity of WMSE		
			$S_{\vartheta_i}^{Tp}$	$S_{\vartheta_i}^{Xp}$	$S_{\vartheta_i}^{CO2}$
r_1[a]	$1.1e-2$	$7.0e-3$	0.6907	0.1382	0.8811
r_2[a]	$-1.9e-3$	$-1.0e-3$	−0.6983	−0.1634	-1.0410
r_3[a]	$1.7e-4$	$2.5e-4$	0.6355	0.3402	**2.1518**
α_{c2}[b]	7.5	3.0	−0.8999	−0.0205	−0.0542
K_{evap}[a]	$9.0e-5$	$1.4e-4$	**4.4291**	**1.2517**	0.1960
E_{CO_2}[a]	$9.0e-3$	$7.0e-4$	−0.0412	0.0125	**1.8932**
G_1	1.0	1.8	**−2.9842**	−0.4426	-0.0721

[a] Literature parameter value taken from Lukasse et al. (2007).
[b] Literature parameter value taken from Kondrashov (2007)

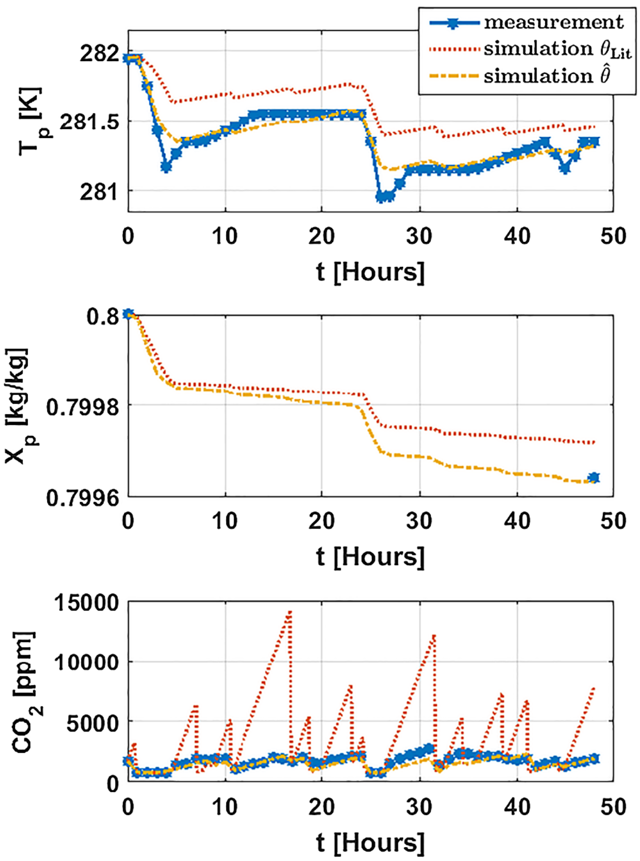

Fig. 4. Calibration results, with the parameter estimates given in Table 3. In all panels, the blue stars indicate the measuring points, the dashed red line is the simulation result using the parameters found in the literature and the dash-dotted yellow line is the model calibration output of the optimal parameter estimates. The top panel shows the temperature of the potato, the middle panel the moisture content of the potato, and the bottom panel the CO_2 concentration. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

From the calibration results, a good fit is obtained for all three states. The simulated potato temperature follows the potato temperature measurements within the sensor accuracy. The potato weight was measured only twice. Hence, the exact intermediate dynamics cannot be verified. However, the moisture loss after 48 h gives a good fit. The CO_2 concentration is in a good range. Only between $t = 28 - 31$ h the concentration is lower compared to the data, but the results are much better than those obtained with the literature parameter values.

Table 3 contains the parameter values found in the literature ($\vartheta_{i,Lit}$), the parameters estimates ($\hat{\vartheta}_i$) with the minimum WMSE, and the sensitivities of the output to the parameters ($S^x_{\vartheta_i}$). The simulations performed with the parameters found in the literature (θ_{Lit}) and the parameter estimates ($\hat{\theta}$) resulted in a WMSE of 56.39 and 1.81, respectively.

Six of the parameter estimates ($\hat{\theta}_i$) are in the same range as the parameter values found in the literature. The estimate of E_{CO_2} is one order lower relative to the value reported in the literature. This result can be ascribed to the fact that another potato variety was used, other growing conditions and probably the usage of sprout inhibitors. The sensitivity results indicate that a change in the parameter estimate of evaporation (K_{evap}) has the greatest influence on the temperature T_p, as well as, on the moisture content X_p. The results also show that the gain parameter G_1 has a considerable influence on T_p. A change in the parameter r_3 and E_{CO_2} have the most influence on the carbon dioxide state (CO_2).

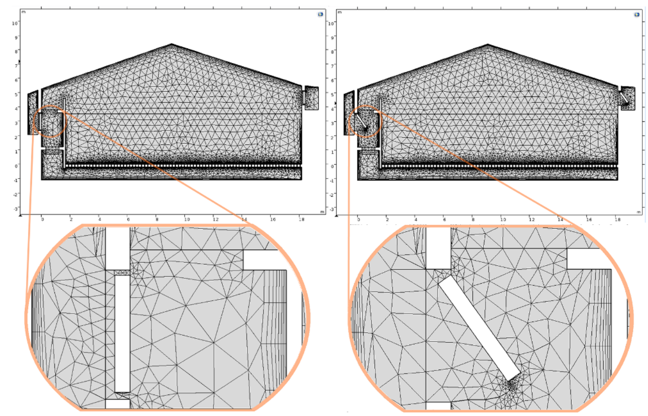

Fig. 5. Mesh configuration of the storage facility. Left panels: initial mesh configuration. Right panels: mesh configuration after some re-meshing steps (t = 26.81 [h]).

3.3. Validation

The validation of the model was performed using data from the same storage facility, storage season, and potato tubers as for the calibration. The only difference is that we took data from a different period of 36 h. For the validation run the complete bulk-storage model was simulated without decompositions.

To perform a dynamic simulation of the complete storage facility, a moving mesh was used to open and close the hatches repeatedly. Having a suitable mesh to solve the storage facility problem is essential. Choosing a mesh that is too coarse will result in inaccurate results. Taking a very fine mesh will lead to long computation times due to the large number of grid points, and is susceptible to errors in the repeated re-meshing of the moving mesh. We used 15466 triangular mesh elements and 2515 boundary elements for the facility (see Fig. 5). The spatial discretisation error (DE) of the mesh for the most important states were estimated by means of a Richardson extrapolation (Roache, 1997). The DE for velocity, temperature, moisture content and the carbon dioxide state were 2.41%, 1.03%, 0.14% and 0.01%, respectively.

As the hatches opens and closes repeatedly during the simulations, the mesh is adapted during the time simulation. In Appendix A, Fig. A.15, nine close-up frames from the first time the hatch opens and closes, between 0.4 and 0.7 [h] of the validation simulation are shown. The frames show that after the first time opening and closing the hatch, a significant smaller mesh around the down part of the hatch is created.

3.3.1. Flow model validation

Fig. 6 provides an impression of the situation with forced and natural convection. The two top panels show the air velocity at two different time instances, and the two bottom panels show the corresponding air temperature profiles. In the forced convection case (left panels), a homogeneous distribution of air and temperature is obtained, as a result of the perforated floor. Including the perforated floor in the model results in longer computational times (small mesh elements), but represents the reality well. In the natural convection case (right panels), there is air circulation due to temperature differences. Both cases show realistic velocity and temperature distributions as conformed in Fig. 7 for the four validation points (see Fig. 1).

The velocity profile of the simulation shows the same behaviour compared to the measurements. The temperature profile above the fan shows a good fit, but after a ventilation action the temperature stays lower compared to the measurements. The roof temperature follows the same trend, but between $t = 6$ and 7.5 [h] a significant higher temperature is obtained in the facility.

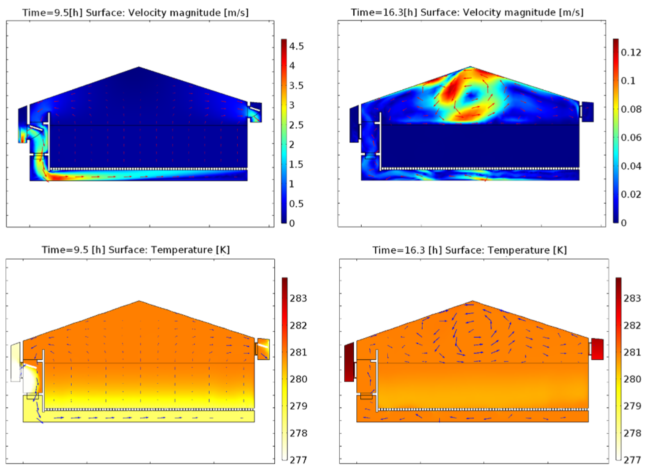

Fig. 6. Validation results. Top left panel: air velocity during forced convection with open hatch. Bottom left panel: air temperature during forced convection with open hatch. Top right panel: air velocity during natural convection with closed hatches. Bottom right panel: air temperature during natural convection with closed hatches.

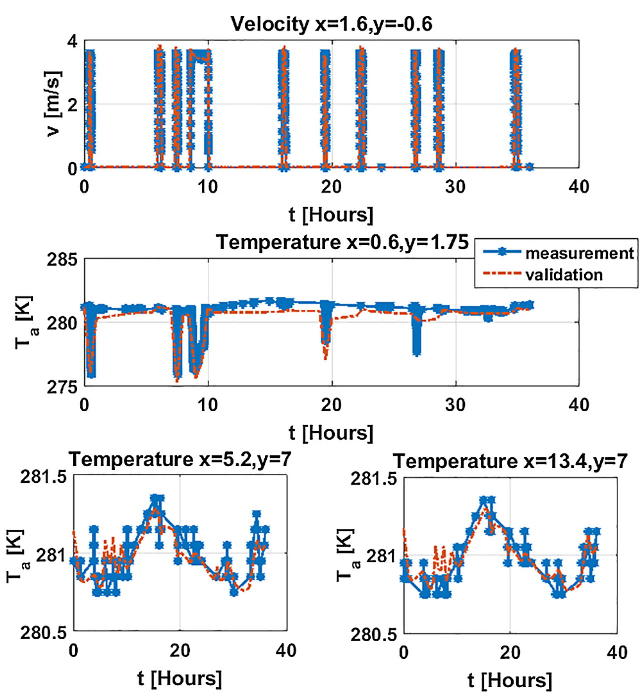

Fig. 7. Validation results. Top left panel: air velocity in air channel. Middle panel: air temperature above the fan. Bottom left panel: air temperature left part under roof. Bottom right panel: air temperature right part under roof.

3.3.2. Product model validation

Fig. 8 shows the validation results of the product model. Likewise as in the calibration, appropriate fits were obtained. The potato temperature (T_p) and moisture content (X_p) are within the sensor accuracy. The carbon dioxide (CO_2) level shows a comparable fit as in the calibration. However, directly after opening the hatches and switching on the forced ventilation, a rapid decrease followed by a rapid increase can be seen in the measurements, but not in the simulation. The simulations and measurements show the same trend after this rapid decrease and increase. In the validation step, the WMSE is equal to 2.48, which is lower than the simulation with literature parameters (WMSE = 56.39), but slightly higher compared with the calibration WMSE of 1.81.

Additionally to the midpoint validation, the potato temperature (T_p) was validated on six other points in the bulk (Fig. 9). Four of the six

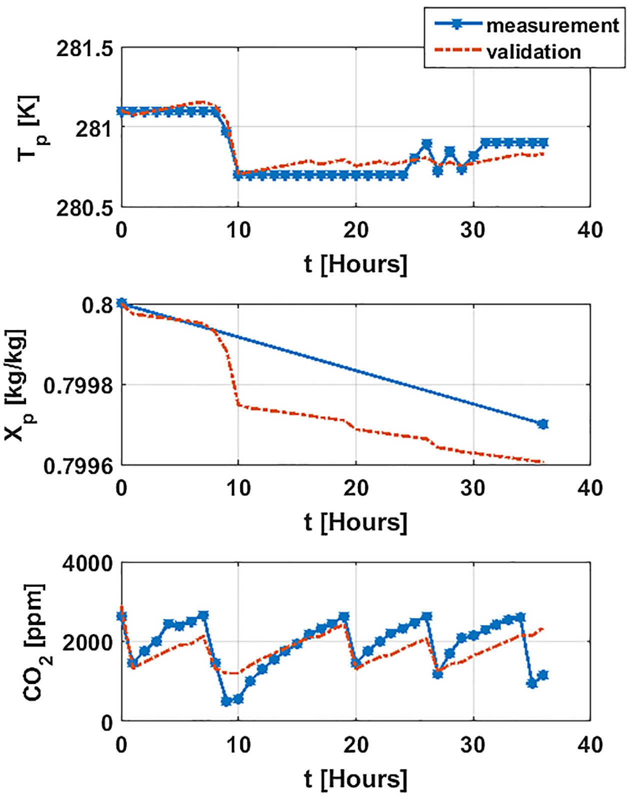

Fig. 8. Validation result of product model. In all panels, the blue stars are the measuring points and the dash-dotted red line is the simulation output. The top panel presents the temperature of the potato, the middle panel the moisture content of the potato, and the bottom panel the CO_2 concentration. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

points, in the top and middle layer of the bulk with coordinates [4.4, 3.0], [15.4, 3.0], [4.4, 1.75] and [15.4, 1.75], show a good fit within the sensor accuracy. For two of the six points, on the bottom layer, on the points [4.4, 0.5] and [15.4, 0.5], a same trend as the measurements are obtained, however, a larger offset is seen. The MSE of the points [4.4, 3.0], [15.4, 3.0], [4.4, 1.75], [15.4, 1.75], [4.4, 0.5] and [15.4, 0.5], are 0.0150, 0.0042, 0.0034, 0.0019, 0.0395, 0.0529 respectively.

3.4. Case study

From the measured data and the validated model simulations, mainly gradients in the vertical direction were obtained for air and product temperature. For the moisture content no significant gradients were obtained, probably due the relative short time simulation. In Fig. 10 the air and product temperatures are shown for t = 36 [h]. The gradients are unwanted, as described in Section 2.6. To minimize the heterogeneity in the bulk, a new ventilation strategy is taken into consideration. In our case study a combination of over- and underpressure is applied to minimise the vertical gradient. A simulation of 10.5 h is performed, using the inlet and outlet data of the validation data set. We show only the results for the forced convective period, i.e. t = 8.5–10.5 [h].

In this case study (case study 1) the pressure profile changes around t = 9.3 [h], from over- to underpressure. The influence on the air temperature profile at time t = 9.5 [h] is shown in Fig. 11. The influence on the potato temperature profiles for 9 different locations is shown in Fig. 12. The vertical gradient at the end of the cooling period are listed in Table 4. As we can conclude from the results the potato temperature gradient is much lower applying over- and underpressure. However, the temperature gradients in horizontal direction are bigger compared to

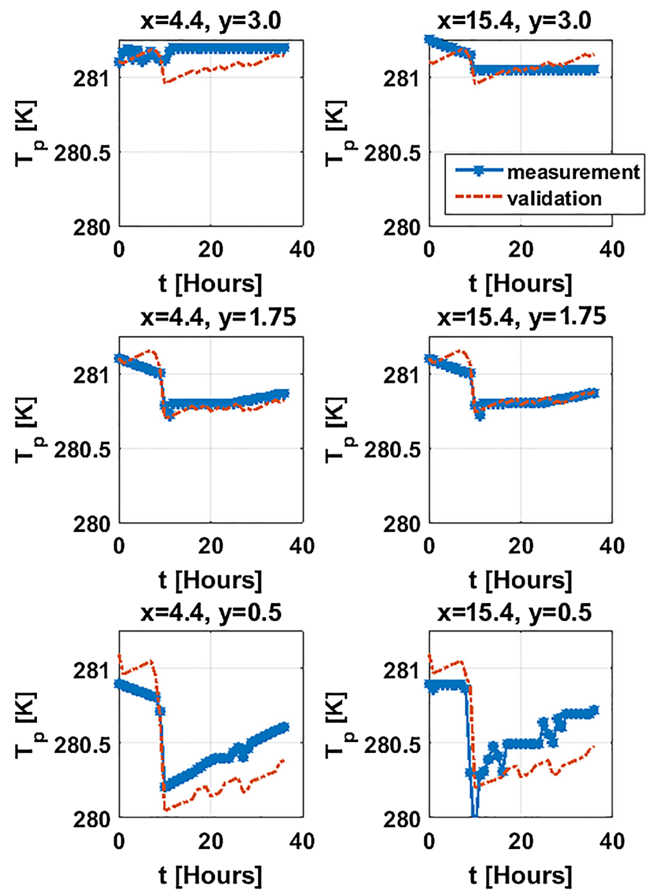

Fig. 9. Validation result of product model. In all panels, the blue stars are the measuring points and the dash-dotted red line is the simulation output. The top panels presents the temperature of the potato at coordinates [4.4, 3.0] and [15.4, 3.0], the middle panels presents the temperature of the potato at coordinates [4.4, 1.75] and [15.4, 1.75], and presents the temperature of the potato at coordinates [4.4, 0.5] and [15.4, 0.5]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

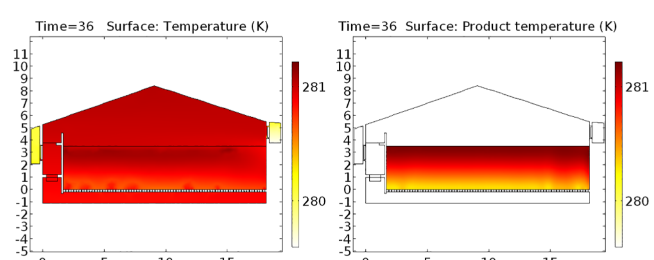

Fig. 10. Air and product temperature profiles for the original simulation for time t = 36 [h].

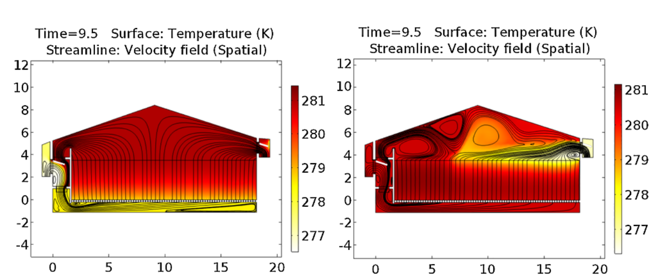

Fig. 11. CFD temperature profiles for two different simulations at time t = 9.5 [h]. Left panel: the original simulation, only over-pressure is applied. Right panel: the simulation with an over- and underpressure applied.

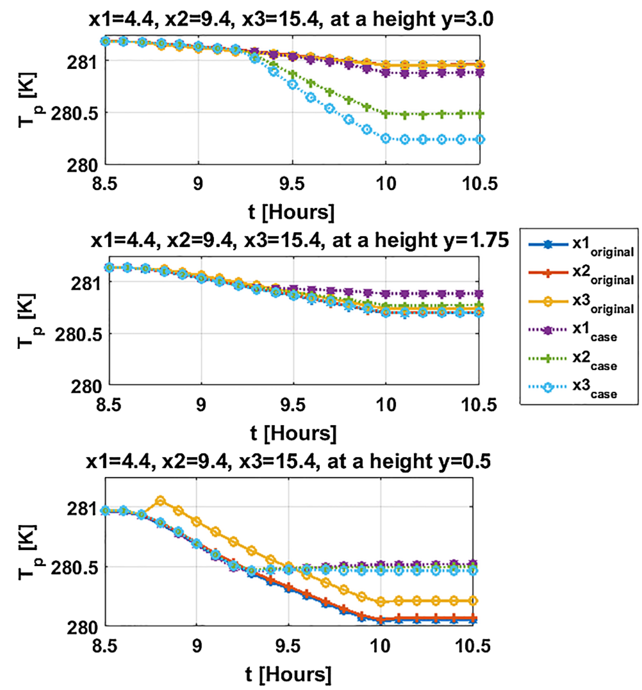

Fig. 12. Product temperature profiles for the original simulation and the case study using over- and underpressure.

Table 4
Product temperature gradients in vertical direction (case study 1).

	Study	x = 4.4	x = 9.4	x = 15.4
$T_p(x, 3.0) - T_p(x, 0.5)$	Original	0.91	0.89	0.75
	Case 1	0.36	−0.01	−0.23

Table 5
Product temperature gradients in horizontal direction (case study 1).

	Study	y = 3.0	y = 1.75	y = 0.5
$T_p(15.4, y) - T_p(4.4, y)$	Original	0	0.05	−0.38
	Case 1	−0.65	−0.19	−0.06

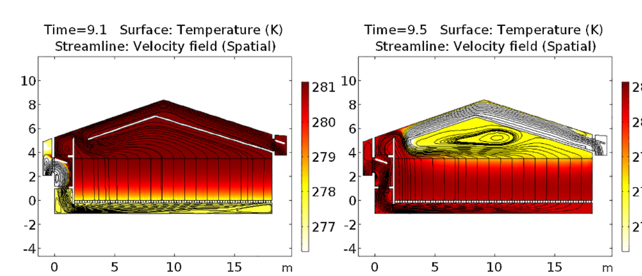

Fig. 13. CFD temperature profiles for an adjusted configuration using an extra air channel near the roof. Left panel: overpressure at t = 9.1 [h]. Right panel underpressure at t = 9.5 [h].

the original simulation, see Table 5. The average temperature in the bulk for the original and the case study simulation at t = 10.5 is 280.59 [K] and 280.60 [K] respectively. The average moisture content for the original and the case study simulation at t = 10.5 is 0.79974 [kg/kg] and 0.79975 [kg/kg] respectively. This implies that on average more or less the same cooling effect is achieved, with smaller gradients in vertical direction and slightly lower weight loss for the case study with over- and underpressure. Thus this result shows that the gradient in vertical direction is reduced, if a combination of over- and

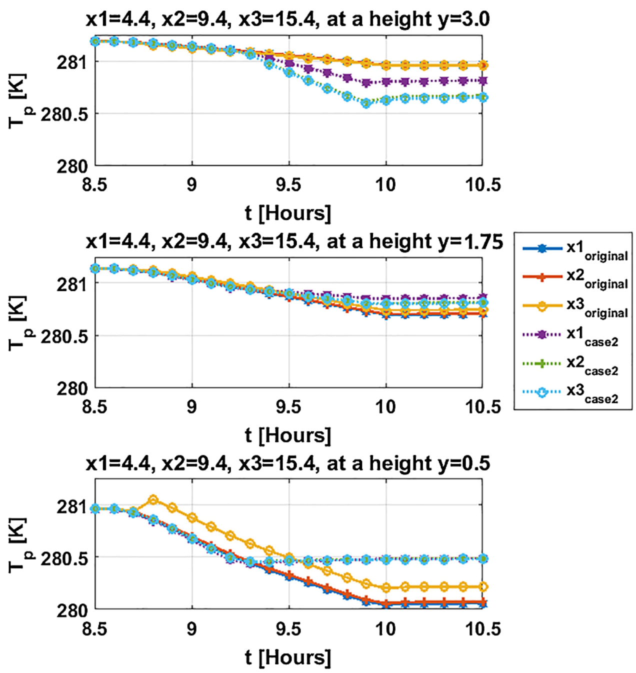

Fig. 14. Product temperature profiles for the original simulation and the case study using over- and underpressure, and an extra air channel near the roof.

Table 6
Product temperature gradients in vertical direction (case study 2).

	Study	$x = 4.4$	$x = 9.4$	$x = 15.4$
$T_p(x, 3.0) - T_p(x, 0.5)$	Original	0.91	0.89	0.75
	Case 2	0.32	0.18	0.17

Table 7
Product temperature gradients in horizontal direction (case study 2).

	Study	$y = 3.0$	$y = 1.75$	$y = 0.5$
$T_p(15.4, y) - T_p(4.4, y)$	Original	0	0.05	−0.38
	Case 2	0.16	0.05	0.01

underpressure is applied. However, the gradients in horizontal direction becomes bigger.

4. Discussion

As described, we have developed a dynamic potato storage model, which we calibrated by decoupling the fast and slow dynamics of the model. For each, a parameter sweep is performed to gain an impression of the feasible parameter region. While this approach is effective in finding feasible parameter vectors for complex models, its drawback is that it generally does not yield optimal parameter estimates.

Consequently, the calibration of the flow model produces a hypercube with width [5, 0.1, 5] of feasible parameter vectors. This calibration result shows that the Darcy-Forchheimer model can be used for the modelling of full-scale storage facility, albeit that the estimates of P_{fan} and c_2 are highly correlated. The product model was also successfully calibrated (see Fig. 4).

Next, we used CFD simulation software as this is a great tool to visualise spatially distributed dynamical behaviour. The dynamic behaviour of the bulk storage facility wasa described in our study by a spatially distributed model with moving hatches and time-varying fan pressure. Simulating the full facility, without decomposition, presented a challenge due to the different time scales in the facility, the relatively small changes in dynamics and the moving mesh. The mesh selection is of great importance for obtaining a satisfying solution. Simulating a full-scale facility relative to simulating a small object automatically means that for the same accuracy, many more mesh elements are needed. To limit memory problems, however, we constrained the mesh of the bulk facility at the price of some acceptable numerical errors and gaining potential numerical instabilities.

The COMSOL CFD simulation tool provides a basis to simulate all relevant dynamics described in this paper. Solving the case with intermittent forced and natural convection in one go while combining the product model with a moving mesh, however, is also challenging. Repeated re-meshing is a good option for simulating a hatch that opens and closes, but there is a limit to the shifting of the mesh elements. For our configuration and mesh choice, the mesh elements could no longer be shifted after 36 h of simulation (see Fig. 8). If we had chosen a smaller mesh, this behaviour would have been occurred even sooner. As a consequence, numerical errors occurred, especially in the CO_2 state. We therefore added numerical diffusion to the diffusion of the carbon dioxide to make the solution stable. This adjustment implies that there are some differences between the calibration and validation models.

The validation of the product model was performed on the same data set of 36 h as we used for the flow model validation. The temperature and moisture states of the potato (T_p and X_p) show differences between measured and predicted states but these are within the sensor accuracy. The spatial distribution of the model is validated using seven potato temperature measurement points. The middle and top part show good results, the lower part of the bulk shows the same trend but not all steep fluctuation seen in the measurements are seen in the simulation. As the correlation of the potato temperature and potato moisture content of the simulation data was found to be 0.89, we assume that the partial distribution found for the temperature distribution also holds for the moisture content, but this needs to be further investigated. The carbon dioxide state (CO_2) shows lower concentrations for three time ranges, but follows the same trend as the measured data.

The mismatch in the carbon dioxide is particularly seen after a forced ventilation action with open hatches. The measurements show a rapid decrease and increase of the carbon dioxide concentration, whereas this behaviour is not observed in the model simulations. The mismatch could be caused by the added numerical diffusion as it changes the dynamic behaviour of carbon dioxide diffusion. After the forced convection, natural convection occurs. Because of the added numerical diffusion, extra carbon dioxide produced by the potatoes is transferred to the air spaces (Ω_{ai} Fig. 1), while in reality this diffusion-driven movement takes more time.

The case study showed that by using an combination of over- and underpressure, instead of only the conventionally applied overpressure, reduces the vertical gradient. However the gradient in horizontal direction becomes bigger, due to the absent of an ideal mixed air stream above the bulk in domain Ω_{a4}. Introducing an extra air channel near the roof will minimize the effect of the non-ideally mixed air. This extra adjustment is simulated in a second case study (case study 2), and the results are shown in Fig. 13. From Fig. 13 it is seen that a more homogeneous air distribution is obtained compared to the case study 1, were underpressure is applied without the air channel at the roof, see right panel Fig. 11. Using the extra air channel leads to a more homogeneous distribution of the product temperature after a cooling period, see Fig. 14. From Fig. 14 we also see that for the bottom part ($y = 0.5$) the temperature is 280.5 [K], in the middle the temperature is around 280.75 [K], and at the top around 280.7 [K]. The product temperature gradients, therefore, are lower compared to the original case and the case study without the additional channel. This is confirmed by the product temperature gradients, see Tables 6 and 7.

The moisture loss did not show significant changes in the case study compared to the conventional case. From practise it is known that for a conventional bulk system most weight losses occur at the bottom layer of the facility, due to the exposure of air with a lower humidity during forced (blowing) ventilation with outside air. When using over- and underpressure it is likely that the bottom part is exposed shorter to dry air, and probably the weight loss will decrease in the bottom part and increase in the top part. This was not confirmed in the simulation, as the simulation time was to short and hence, further investigation on a complete storage season is needed.

In Appendix A, Fig. A.16 shows the three cases (for existing facilities) at time t = 10.5 [h]. From Fig. A.16 we see that the potato temperature distribution in the case study 2 is more homogeneous distributed compared to the other two cases. However, in the left corner and in the middle still some warmer areas and at the bottom a slightly colder area can be seen. For improved air mixing for over- and underpressure in new facilities, the right outlet hatch could be placed at the left side, such that the fan is in between two hatches.

5. Conclusion

In this paper, we have proposed a modelling approach and a spatially distributed air-product model for an industrial full-scale potato bulk storage facility with dynamic control elements. The model was successfully calibrated after decoupling the fast (air-related) and slow (product-related) dynamics. The calibrated CFD model was also successfully validated with full-scale bulk storage data.

Since this model contains all relevant dynamics and control components (fan and hatches), it can be used to obtain more insight in the relation between storage climate, quality development and storage configuration.

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Appendix A. Additional figures

Figs. A.15 and A.16.

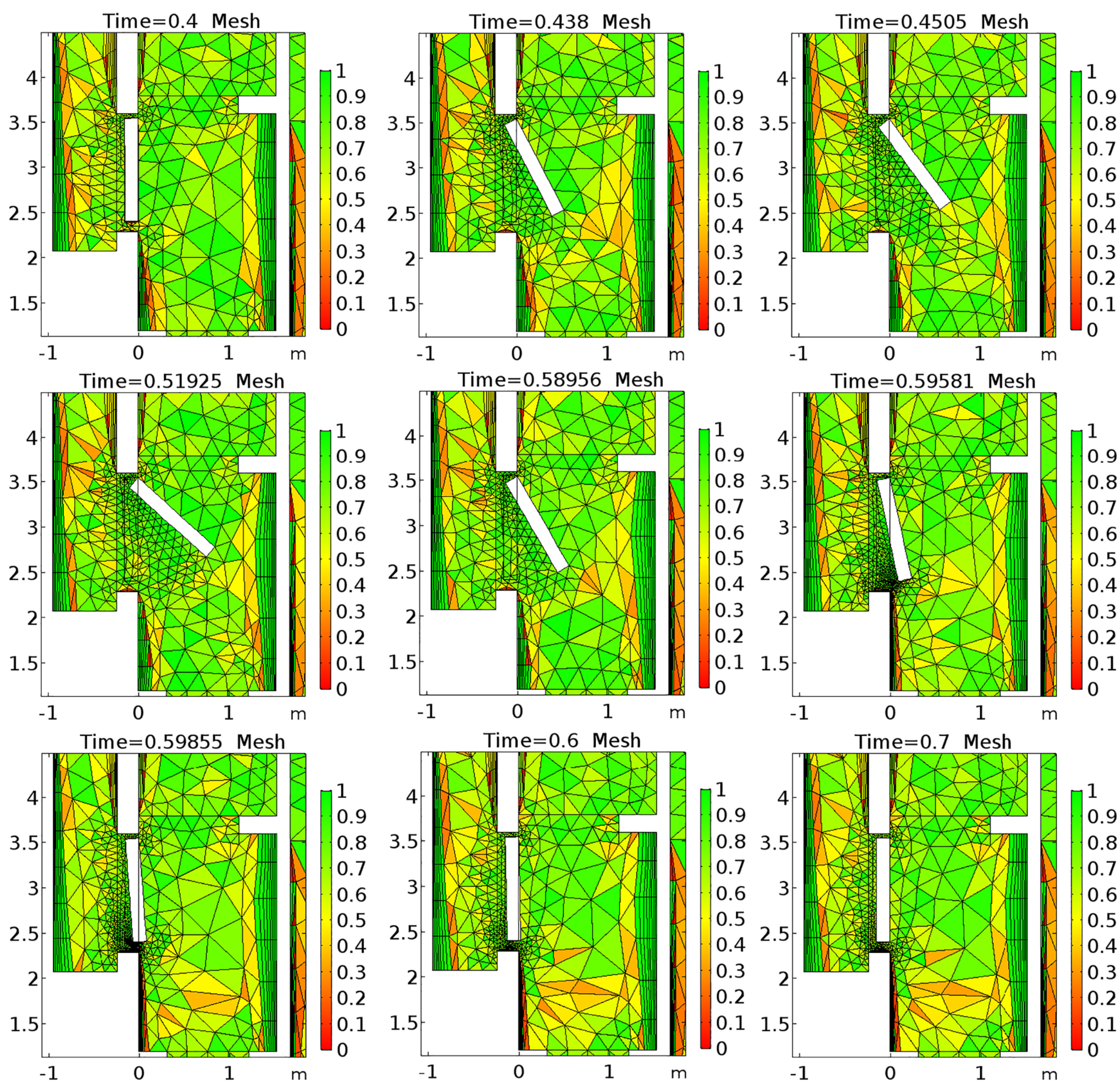

Fig. A.15. Dynamic implementation of opening and closing the inlet hatch, using a moving mesh. Results shown for time: t = 0.4 [h] to 0.7 [h]. The colours indicate the equiangular skewness.

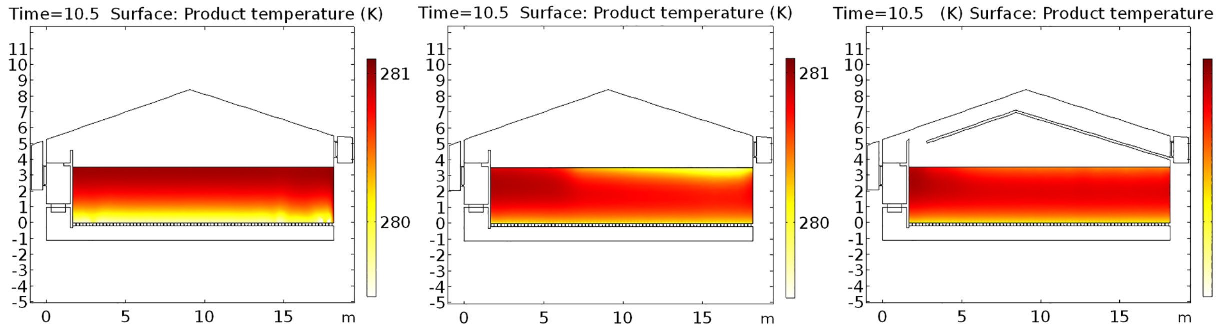

Fig. A.16. Product temperature at time t = 10.5 [h]. Temperature distribution after forced convection of the original case (left panel), the case study using over- and underpressure (middle panel) and case study 2 with an over- and underpressure and an extra air channel near the roof (right panel).

Appendix B. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.compag.2018.12.033.

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